HelioVolta:

HelioVolta — Founded on engineering expertise, we give our clients clarity and confidence in their assets through rigorous inspections, technical due diligence, and oversight. Our team of in-house engineers ensures that your energy systems are built right, perform reliably, and meet the highest standards at every stage of the project lifecycle.

We've supported over 1,500 projects totaling more than 20 GW, backed by a team with 50+ years of combined industry experience. Our NABCEP-certified inspectors and in-house engineers bring deep technical insight to every site. All fieldwork is managed through our SolarGrade platform to ensure consistent, professional reporting from start to finish.

HelioVolta was founded by James Nagel, PE, and David Penalva, two NABCEP-certified PV system inspectors and field engineers. They observed the rising trend of quality issues in renewable energy projects since joining the solar industry in the 2010s.

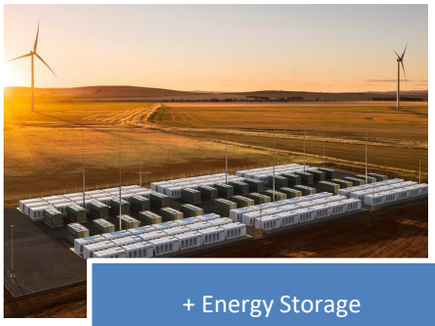
They believed rapid industry growth, inconsistent training, and evolving technology led to preventable safety and reliability failures. Their two-part solution combined independent technical services and SolarGrade, a purpose-built software platform for field operations and asset management.

Areas Of Expertise:

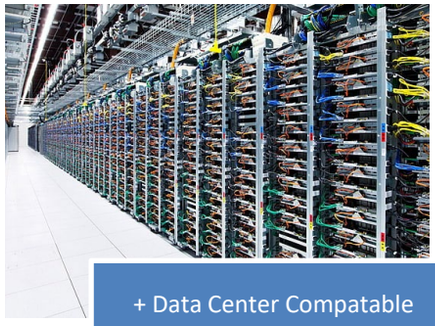
Ground Mount

Commercail Rooftop

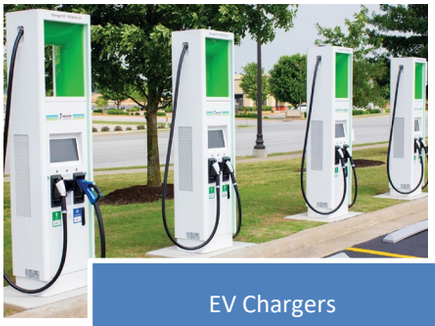
Cannopy



+ Energy Storage



+ Data Center Compatable



EV Chargers

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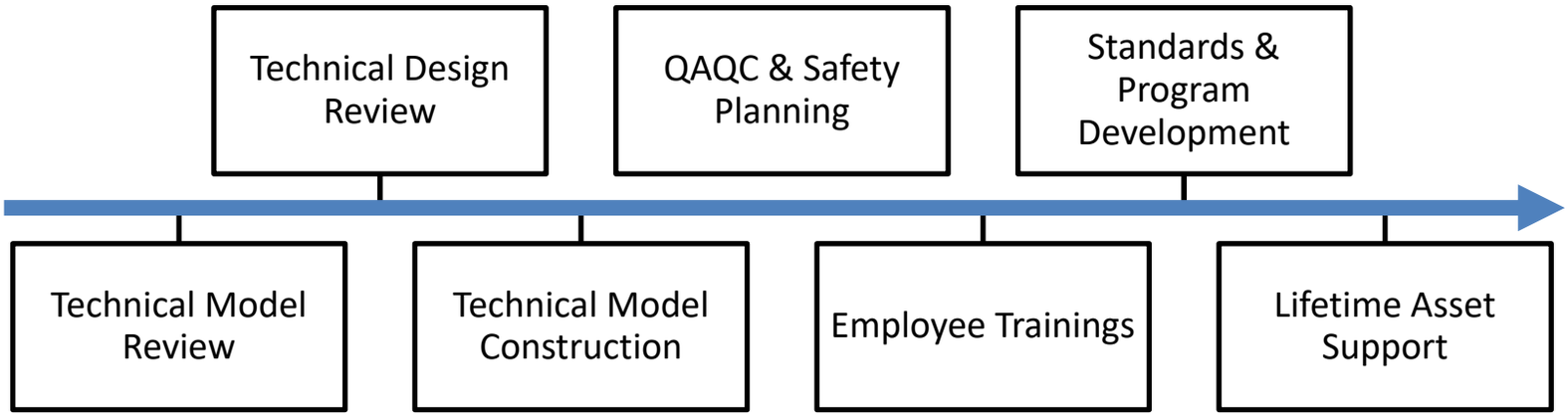
Pre - Construction Services

HelioVolta helps clients plan their projects to ensure the creation of quality installations. Our expert assessments of project plans, components, performance models, and timelines. The decades of experience in our team allow for reviews to identify risks, ensure compliance with codes and regulations, and validate that projects align with safety, quality, and financial targets.



Technical Model Review:

HelioVolta's in-house engineers perform in-depth design and model reviews to validate engineering plans, performance models, component selections, and procurement specifications. The team identifies code compliance issues, safety or reliability risks, and financial red flags before construction begins.

Technical Design Review:

The HelioVolta engineering team re-examines engineering plans and models to ensure safety, code compliance, and alignment with industry best practices.

Technical Model Construction:

HelioVolta collaborates with clients to assess asset-specific challenges. From soiling patterns to insolation modeling, internal analysts guide risk assessment and help determine which modeling approaches provide the most value based on client needs.

QAQC & Safety Planning:

HelioVolta implements ongoing quality assurance processes, code compliance checks, safety audits, risk-based punch lists, and safety planning frameworks. Using the SolarGrade platform, the team enforces standards and documents construction anomalies in real time.

Employee Trainings:

HelioVolta delivers hands-on training workshops and technical guidance to field teams focused on best practices in installation, equipment handling, wire management, construction quality tracking and verification practices.

Standards & Program Development:

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Lifetime Asset Support:

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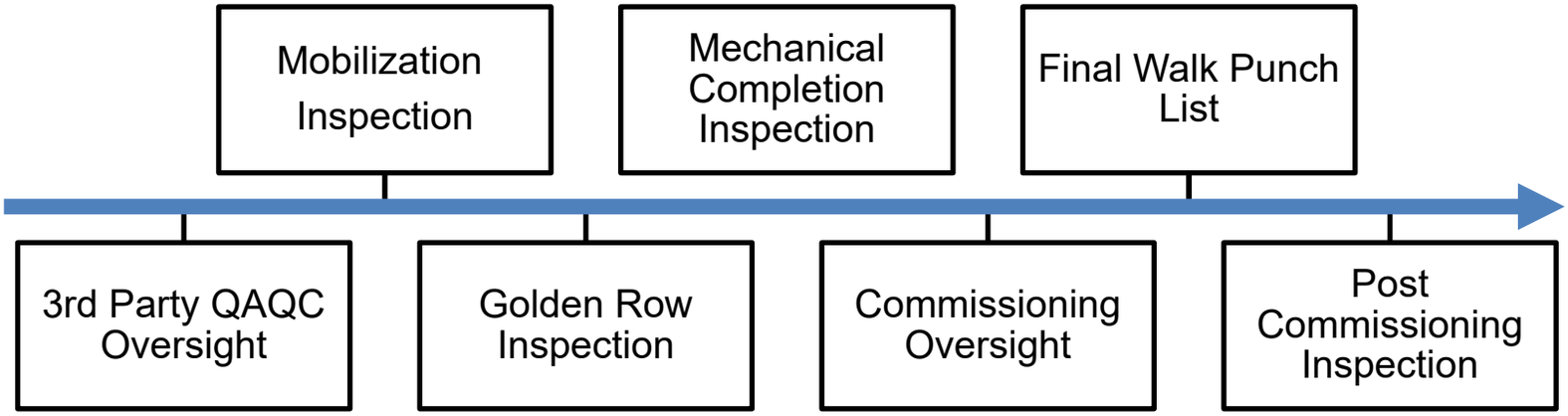
Active Construction Services

HelioVolta works to identify, resolve, and prevent issues during construction. Our services establish and enforce quality standards from day one, reducing the risk of failure and underperformance during operations.



3rd Party QAQC Oversight:

HelioVolta provides independent quality assurance and quality control (QAQC) oversight during construction to identify workmanship issues, safety risks, and installation defects. This third-party verification ensures alignment with industry standards, project specifications, and contractual requirements.

Mobilization Inspection:

HelioVolta conducts inspections at the onset of construction to verify proper mobilization of equipment, materials, and workforce. The team evaluates site readiness, safety protocols, staging logistics, and delivery quality of key components.

Golden Row Inspection:

HelioVolta inspects the initial row of installed modules and racking, referred to as the “Golden Row”, to confirm installation methods and ensure construction teams are following best practices before full-scale deployment begins.

Mechanical Completion Inspection (Cold):

At mechanical completion, HelioVolta performs a detailed inspection to confirm all equipment is installed per design and ready for commissioning. The team checks racking, modules, wiring, inverters, and other balance-of-system components for completeness and quality.

Commissioning Oversight:

During commissioning, HelioVolta provides oversight to ensure testing procedures are executed properly and that the system performs as intended. This includes observing functional tests, verifying inverter and system start-up protocols, and confirming system safety and operability.

Post Commissioning Inspection (Hot):

Following system commissioning, HelioVolta conducts a thorough inspection to identify latent issues not during energization. This includes safety checks, performance validation, and documentation of any deficiencies that may impact long-term operation.

Final Walk Punch List:

HelioVolta performs a final walk-through to generate or validate the project punch list. This inspection ensures all prior issues have been addressed, outstanding corrective actions are documented, and the system is ready for closeout or handoff to operations.

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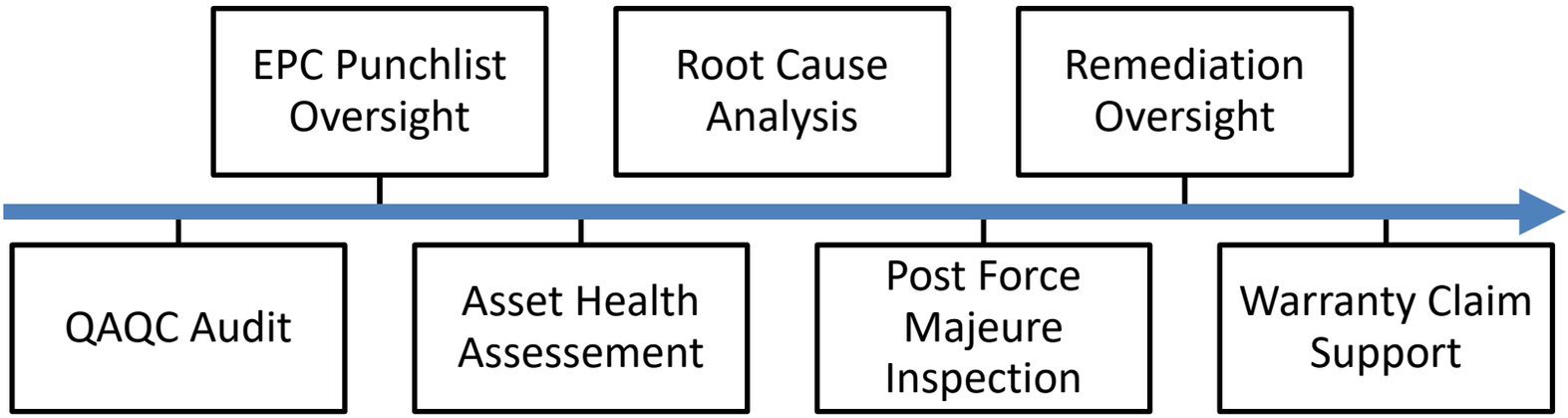
Post Construction Services

HelioVolta's team of in-house engineers conducts on-site inspections of operating assets using advanced visual, thermal, electrical, and lab testing to detect safety and quality risks early. Our statistically significant sampling maximizes insights while minimizing risks and disruptions to ongoing operations.



QAQC Audit:

HelioVolta performs comprehensive quality assurance and quality control (QAQC) audits to evaluate the installation quality, safety compliance, and workmanship of PV and energy storage systems. These audits are conducted at key construction milestones or during operations to detect and document issues before they escalate into performance or safety failures.

EPC Punchlist Oversight:

HelioVolta provides oversight and validation of punch list items generated during EPC project closeout. The team ensures identified deficiencies are addressed properly and verifies that the system meets contractual, safety, and performance standards before final acceptance.

Asset Health Assessment:

HelioVolta conducts in-field health assessments to evaluate system performance, identify early signs of degradation or failure, and provide clarity of overall system safety. These assessments support proactive defect remediations, risk mitigation, and overall asset understanding.

Root Cause Analysis:

In response to system underperformance or equipment failure, HelioVolta performs root cause analysis to identify the origin of technical issues. This process involves design and log reviews, field inspections, and technical diagnostics to determine core issues in order to construct a comprehensive remediation plan.

Post Force Majeure Inspection:

Following natural disasters or unexpected events such as fire, hail, wind, or flooding, HelioVolta conducts post-event inspections to assess assets to evaluate mechanical and electrical safety and system damages. On site engineering will support documentation of event effects and recovery planning with a clear path to site remediation.

Warranty Claim Support:

HelioVolta supports clients in preparing and substantiating warranty claims by conducting independent inspections, documenting failures, and organizing evidence packages. Their reporting helps ensure claims are technically sound and meet manufacturer, EPC, owner, and other warranty requirements.

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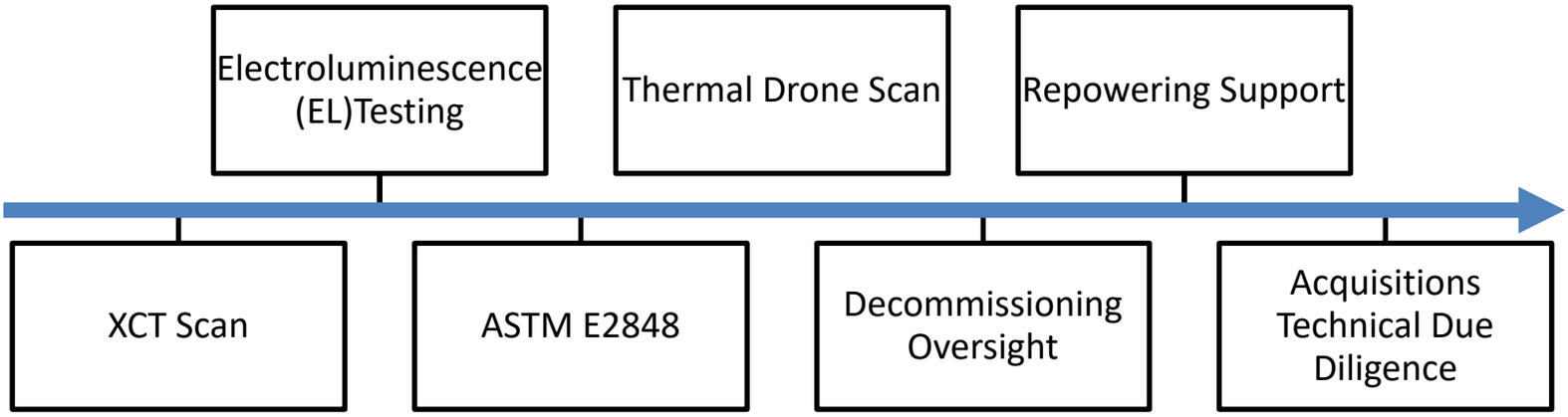
Diagnostics

HelioVolta supports individual diagnostics and in-field assessments to help clients better address individual needs.



Electroluminescence Testing:

HelioVolta uses electroluminescence (EL) imaging to identify microcracks, cell fractures, and other internal module defects not visible through standard inspection methods. This technique supports warranty evaluations, degradation analysis, and root cause investigations.

ASTM E2848:

HelioVolta applies ASTM E2848 testing protocols to validate long-term energy production of PV systems. This standard provides a consistent, third-party framework for performance verification, often used during asset handovers or to support performance guarantee claims.

XCT Scan:

HelioVolta utilizes X-ray computed tomography (XCT) to examine the internal structure of PV system components. This non-destructive diagnostic technique is especially effective for identifying defects in compact or sealed elements, such as connectors and junction boxes.

Thermal Drone Scan:

HelioVolta conducts thermal drone scans to identify hotspots, module faults, and electrical imbalances across large-scale solar arrays. This aerial method provides efficient, high-resolution thermal data to support preventative maintenance and asset health assessments.

Repowering Support:

HelioVolta supports repowering initiatives by evaluating aging system components, identifying underperformance drivers, and recommending technical upgrades. Services include field inspections, design reviews, and performance benchmarking to extend asset life and improve energy yield.

Decommissioning Oversight:

HelioVolta provides engineering oversight and QA/QC support during system decommissioning. This includes verifying safe equipment removal, environmental compliance, and asset condition documentation for reuse, resale, or recycling.

Acquisitions Technical Due Diligence:

HelioVolta performs technical due diligence to support investors, developers, and acquirers during solar and storage asset transactions. Services include document reviews, site inspections, risk identification, and performance modeling to validate asset value and inform acquisition decisions.

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